

Scalability & Future Plan

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local deployment only	Limited accessibility	High
2	Single machine execution	Limited concurrent users	Medium
3	No user authentication	Limited security features	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local users only	Deploy to cloud for multiple users
2	Data Storage	Local dataset	Cloud database integration
3	Performance	Local ML prediction	Model optimization & caching
4	Security	Basic validation	User authentication & HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Cloud Deployment	1–2 Months	Remote accessibility
Phase 2	Weather API Integration	2–3 Months	Better recommendations
Phase 3	Fertilizer Recommendation	3–4 Months	Improved farming support
Phase 4	Mobile Application	4–6 Months	Wider user adoption